CHERI FAVEL

Chemical Process Engineer (MEng) | Process Development & R&D

Abu Dhabi, UAE | South African | UAE Resident Visa | Availability: Immediate
cherifavel@gmail.com | +971 58 584 5473
linkedin.com/in/cheri-favel | researchgate.net/profile/Cheri-Favel | Engineering Portfolio



PROFILE

MEng Chemical Engineer with industry and postgraduate research experience in hydrometallurgical process development, electrochemical metal recovery and industrial water treatment. Experience includes laboratory investigations, mass and energy balances, process calculations, PFD/P&ID preparation and equipment-sizing support. Published research on mineral-leaching kinetics, with project experience across cobalt, manganese, zinc and gold recovery. Seeking UAE process engineering, applications or industrial R&D opportunities.

CORE COMPETENCIES

- **Process engineering:** process design support, mass/energy balances, process calculations, equipment-sizing support, PFDs/P&IDs, design bases and specifications, CAPEX/OPEX support.
- **Hydrometallurgy / electrochemistry:** mineral leaching, electrowinning, metal recovery, selective precipitation, manganese processing and reactor testing.
- **Water / environmental:** industrial tailings-water treatment, Fe/Zn/Mn removal, solid–liquid separation, water chemistry; familiarity with UF/NF/RO fundamentals.
- **Laboratory / data:** ICP-OES, XRF, XRD, FTIR, SEM-EDS, BET, TGA; experimental design, kinetic modelling and data interpretation.
- **Software:** advanced Excel (VBA, Power Query, PivotTables), Visio, Python; Aspen Plus/HYSYS foundational exposure.

ENGINEERING EXPERIENCE

Process & R&D Engineer

Mar 2021 – Jul 2022

Free Radical Process Design (FRPD) | South Africa

- Conducted laboratory and engineering investigations to evaluate hydrometallurgical, electrochemical and industrial water-treatment processes.
- Prepared mass and energy balances, process calculations, PFDs/P&IDs, design bases, specifications and technical reports; supported unit design and equipment sizing.
- Planned and coordinated client laboratory work, interpreted test data and prepared engineering recommendations; reviewed supplier quotations.

SELECTED INDUSTRIAL PROJECTS

- **Tailings-water treatment, Peru:** Evaluated oxidation, precipitation and electrolysis routes for Fe, Zn and Mn removal/recovery, assessing treatment performance and feasibility.
- **Continuous cobalt electrowinning:** Contributed to process development and technical evaluation of a continuous electrochemical cobalt-recovery system.
- **Manganese processing:** Investigated selective precipitation and electrochemical beneficiation routes; supported manganese sulphate monohydrate process development.
- **Gold recovery reactor:** Investigated voltage, current density, flow and reaction time during lab-scale reactor development for residual gold recovery from refinery eluate.
- **Rotating-cathode zinc removal:** Evaluated an electrochemical zinc-removal system for tailings water and considered scale-up requirements for mobile-plant application.
- **Additional investigations:** Fe/Al precipitation from leach liquor; Ni-MH battery metal-recovery feasibility; neodymium plant CAPEX/OPEX analysis; galvanising-process review.

Master's Research in Chemical Engineering

Jan 2020 – Dec 2021

University of Pretoria | South Africa

- MEng research - phlogopite leaching: Designed nitric-acid leaching experiments, modelled kinetics and characterised feed and residues; findings published in *Clays and Clay Minerals*.
- Independently planned and conducted laboratory research, including experimental methodology, kinetic modelling, analytical characterisation and interpretation of results.
- Maintained experimental records, supervised research assistants and communicated findings through technical reports and peer-reviewed publication.

ADDITIONAL PROFESSIONAL EXPERIENCE

Business Setup & Marketing Consultant
Apr 2025 – Present

Cresco Consulting FZCO | UAE

- Coordinate business setup documentation, maintain records and support client communications and marketing activities.

Vessel Operations & Engineering Support
May 2023 – Apr 2025

Research Vessel Rosa | United States & Bahamas

- Assisted with vessel equipment checks and troubleshooting involving pumps, plumbing and RO watermakers; followed marine safety procedures; organised technical drawings and maintenance documentation and supported procurement.

Academic Journal Editor
Nov 2022 – Apr 2023

CACTUS Communications | South Africa

- Edited scientific manuscripts for technical clarity, consistency and publisher formatting requirements.

Teaching Assistant (Part-Time)
Feb 2019 – Jun 2021

University of Pretoria | South Africa

- Supported laboratory sessions and tutoring in particle technology, process kinetics and engineering practice/costing; supervised laboratory work, assessed students and provided technical feedback.

Laboratory Intern (Vacation Work)
Dec 2017 – Jan 2019

University of Pretoria | South Africa

- Completed vacation placements totalling 12 weeks in mineral leaching and fertiliser production, including experimental optimisation and analytical testing.

EDUCATION

MEng Chemical Engineering

University of Pretoria, South Africa | 2021

- Dissertation on nitric-acid leaching kinetics of phlogopite and solid-residue characterisation; awarded a postgraduate research bursary.

BEngHons Chemical Engineering (Distinction)

University of Pretoria, South Africa | 2020

- Recipient of Academic Honorary Colours.

BEng Chemical Engineering

University of Pretoria, South Africa | 2019

- Environmental Engineering specialisation including water/effluent treatment, bioreactors, air pollution, waste remediation and environmental regulation.

PROFESSIONAL DEVELOPMENT

- Occupational Safety and Health: Hazard Communication – LinkedIn Learning | Feb 2026
- Project Management – Energy and Water Projects – PMI / LinkedIn Learning | Feb 2026
- LEED Green Associate Exam Preparation - LinkedIn Learning | Sep 2026
- Operations Management Professional Certificate – Jack Welch Management Institute / LinkedIn Learning | In Progress (2026)

LANGUAGES

- English (native), Afrikaans (proficient), Hebrew (basic)